



LIVE BETA

PLATFORM ARCHITECTURE · CAPABILITIES · VERIFICATION

Anil Kumar Mandal · FAU CEGE · September 2026

12
containerised services
one public entry point

223
API endpoints
20 routers · 6 live sockets

79,361
source lines
Python & TypeScript

3
hydrologic engines
1D, 2D, coupled — cross-validated

20
users, concurrent
3–5 ms median response

01 Three pillars over one data fabric

Drone capture → model input → simulation → back to the catalogue, without leaving the browser.

Data Catalog

DISCOVER · PUBLISH · LICENSE

STAC catalogue in PostGIS — spatial and temporal search, map preview of every item

Natural-language search — plain English becomes keywords, a place, a radius, a date range

ISO 19115 records — formal metadata with XML export for every published dataset

Request-access marketplace — owners approve; approved users hold a durable entitlement

Live hazard overlays — FEMA flood hazard layer, weather radar, river-gauge alerts

PyWMP-WEB

MODEL · SIMULATE · CALIBRATE

Keyless data assembly — draw an area of interest, fetch national elevation, land cover, soils, hydrography

1D simulation — HEC-HMS hydrology with Cascade 2001 storage routing: pumps, gated spillways, drop inlets

2D and coupled — GPU rain-on-mesh depth grids; 1D network inflows injected into the same mesh

Calibration — automatic optimisation against USGS gauge flow, scored NSE / KGE / RMSE

AeroFuse

BUILT

CAPTURE · RECONSTRUCT · DETECT

Photogrammetric reconstruction — drone imagery to a georeferenced 3D point cloud, permissively licensed

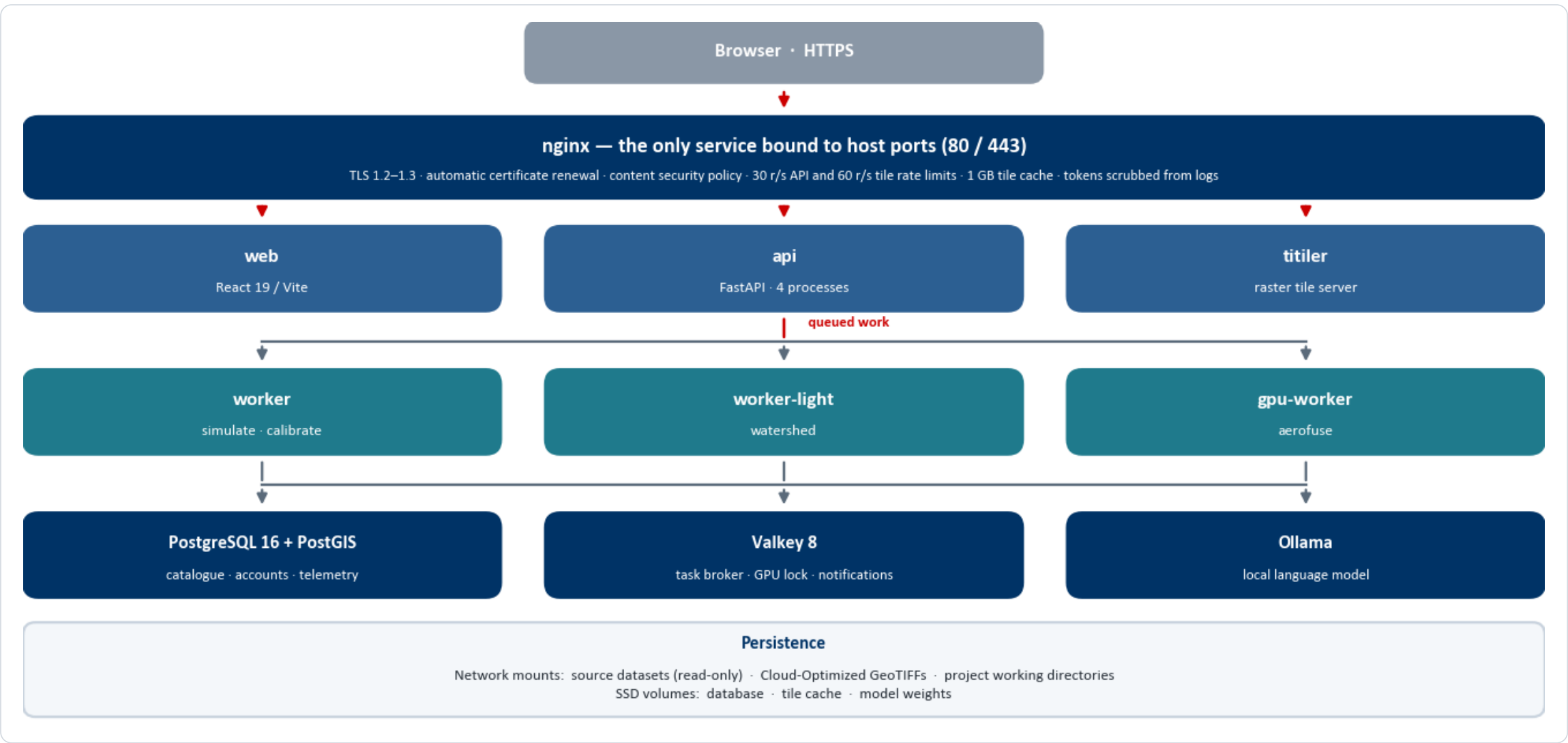
Survey products — surface model, terrain model, orthophoto, classified point cloud

Damage assessment — compare two survey dates, grade each building footprint by height change

Verified end to end on a recorded run (6 images → 4,230-point cloud) but switched off by configuration flag. Enabling it is a release decision, not remaining work.

02 Architecture and deployment

12 containers, one GPU server



Heavy work is queued to three worker tiers, so a simulation cannot block the interactive API. Every run writes a durable record — configuration, log, outcome — so a result traces back to its exact inputs.

03 Technology stack

pinned versions

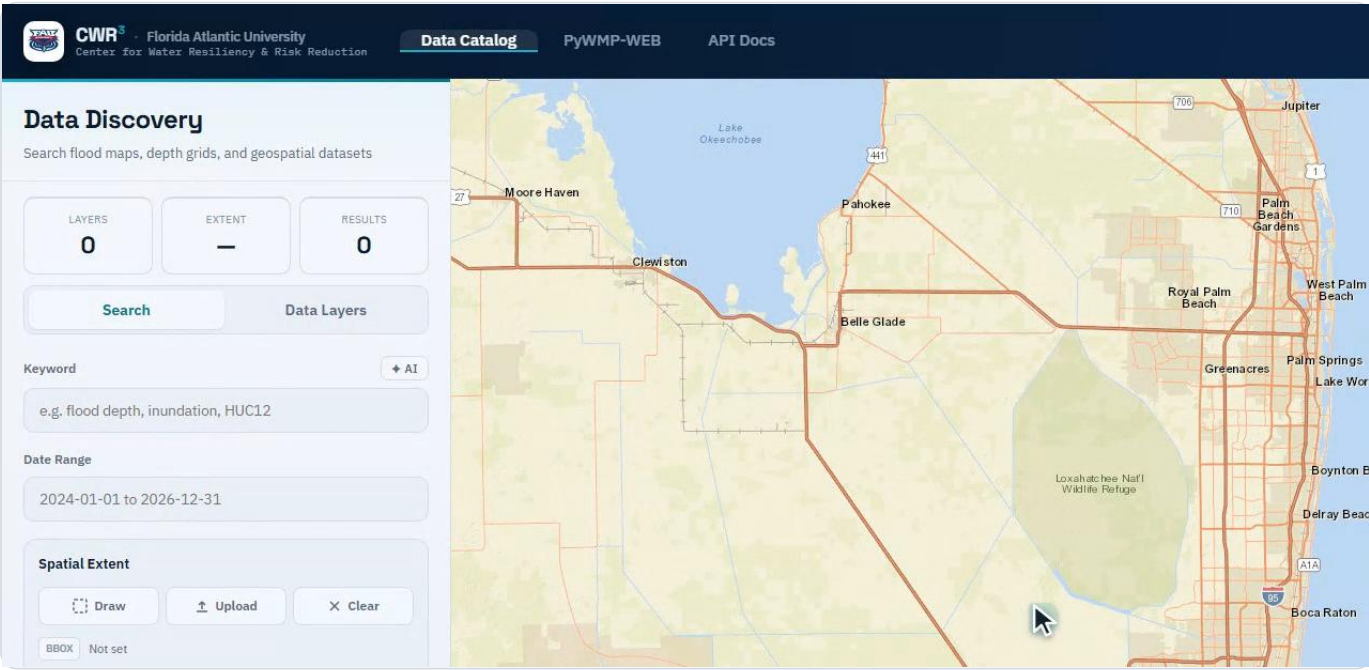
React + TypeScript, Vite
MapLibre GL JS · Chart.js · three.js
FastAPI on Python 3.12
Celery + Valkey broker
PostgreSQL / PostGIS / pgSTAC
rasterio · Fiona · Shapely · PyPROJ
TiTiler tile server
nginx, auto certificate renewal
Ollama, locally hosted
Docker Compose

19.1.0 / 6.3.2
5.3.0 / 4.4.7 / 0.169
0.115.12
5.4.0
16 / 3.4
1.4.3 / 1.10.1 / 2.0.6
0.18.5
1.27
pinned by digest
12 app services

Access is narrow by construction: FAU-email registration with administrator approval, permission checked per resource, restricted downloads gated by an owner-granted entitlement. The system fails closed — it refuses to start on placeholder credentials and runs notebook code in an isolated process with no network.

04 The platform in use

Blackwater Creek HUC-12 · NOAA Atlas 14 100-year 24-hour storm · live beta, September 2026



Discover
Keyword, date and drawn-extent search across the STAC catalogue, every item previewed on the map.

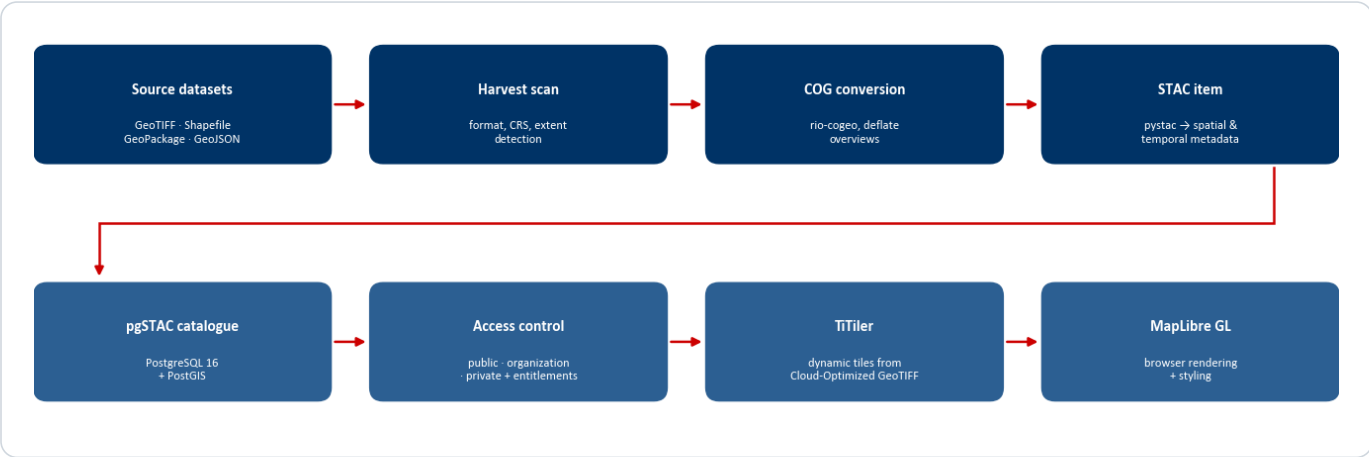


Model
NLCD land cover over 3DEP terrain — the Manning's n and curve-number basis for the 2D mesh.



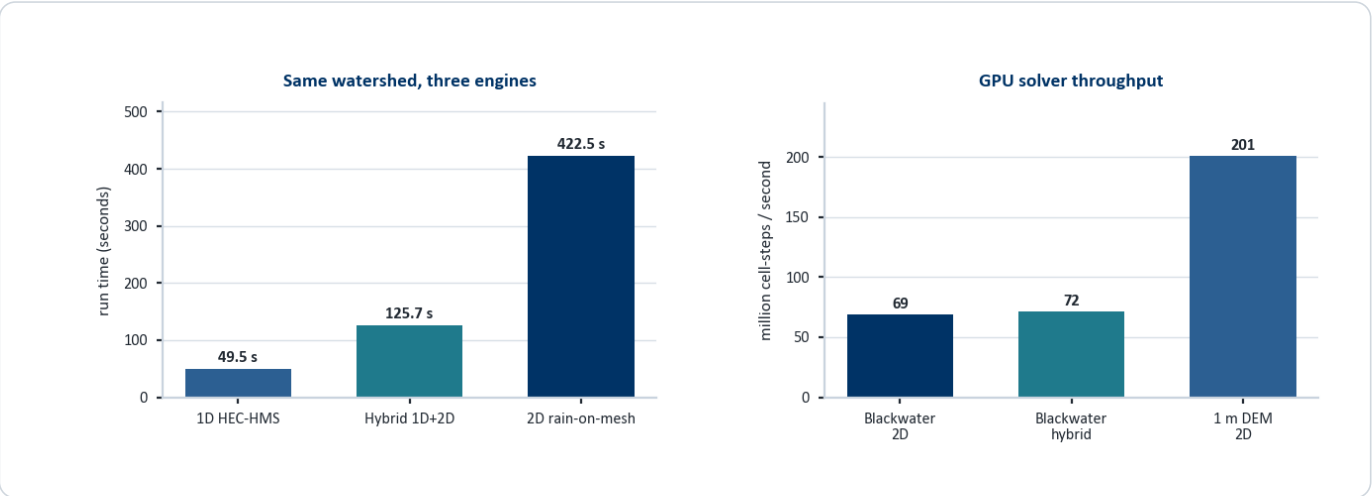
Deliver
PyPilot re-runs the solver on request and answers from the project's real state — on a locally hosted model.

05 Ingestion and discovery



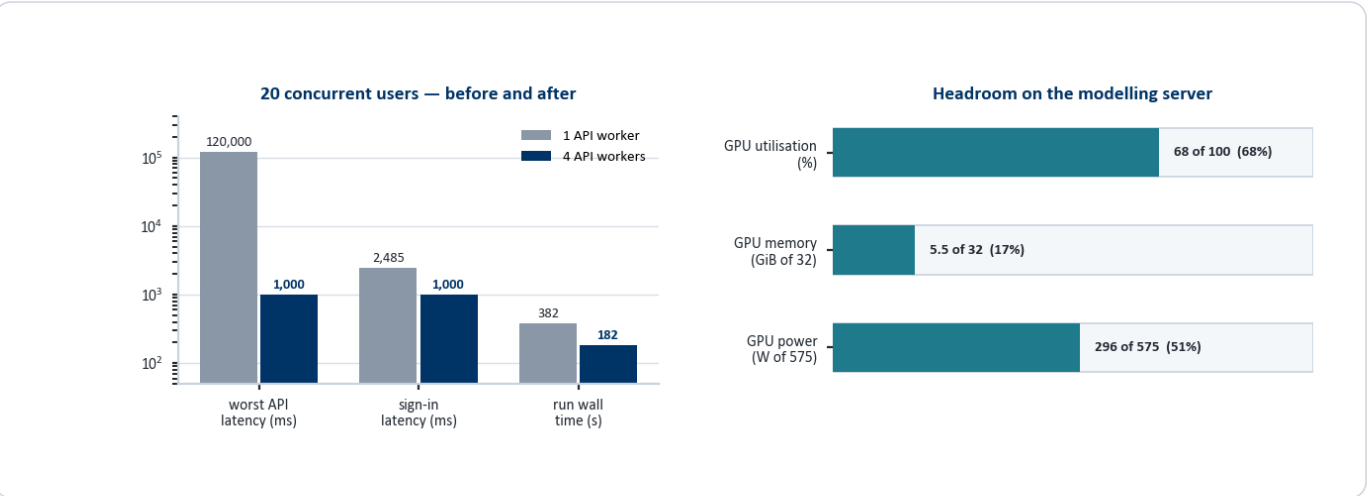
Converted once, then served as tiles — a multi-gigabyte raster is browsed without downloading it.

06 Three engines



100-year 24-hour storm of 10.8 in: peak ≈ 26,260 cfs (1D); max depth 4.77 m over 206.4 km² (2D); the coupled run reproduces the 2D result.

07 20 concurrent users



Four API processes removed the head-of-line block: 4,780 requests at a 3–5 ms median, GPU memory at 17 % of 32 GiB.

08 Where each workflow stands — and the ask

✓ Ready · Building · Team ask

✓ **Feature completeness**
1D, 2D and coupled engines cross-validated

🔑 **Regulatory acceptance**
Technical basis set; formal agency acceptance in progress

✓ **Data interoperability**
Keyless elevation, soils, land cover, rainfall

🔑 **Calibration credibility**
Gauge calibration live; published benchmarks to follow

✓ **Performance and scale**
Multi-process API, GPU compute, measured headroom

🔑 **Maintenance and continuity**
Release cadence and governance need a named team

Release model — confirm the licensing approach through FAU's Office of Technology Development.

Registration — register CWR³Data as a single copyright work.

Team and governance — a named CWR³ team with roles, IP handling and continuity.

Enable the third pillar — agree the release gate for switching AeroFuse on.

TARGET FIRST PUBLIC RELEASE

15–20 Nov 2026